ITERATION 50 / UNIVERSITY RESOURCE ALLOCATION

Repository Maintenance and Regression Assurance

Objective: remove verified obsolete code, centralize proven duplication and reduce calendar database work without changing approved allocation, privacy or lifecycle behavior.

Flask services + PostgreSQL → EngineGateway / JSON → Existing C++17 engine

Existing authority boundaries are preserved; serialization is not allocation logic.

Focused improvements

Component	Maintenance decision and preserved behavior
Presentation and guards	Remove retired selectors, two unused imports, an unreachable exam fallback and an unread snapshot binding. Existing subtype validation and engine evaluation stay active.
C++ test runner	Use an owned temporary build directory with checked-path cleanup on success or failure. Preserve compiler/test exit status and both isolated CLI smokes.
BestFitSelection	One staged-choice capacity selector serves both best-fit exam strategies. Keep distinct room indices, minimum capacity, minimum room count, deterministic ties and shared residual-capacity semantics.
EngineGateway	Share only the five non-legacy operation-v1 response headers. Keep body validation, exact operation errors, legacy allocation-v1 behavior and executable freshness checks.
legacy_intervals	Filter busy slots and active-run requester/committee allocations in SQL; fetch only relevant bindings and periods. Keep private titles, all matching allocations and explicit date-resolution semantics.
Failure-path coverage	Drain process results before joining; clean up owned workers and queues on failure. Add timeout/startup-failure regression guards without relaxing booking assertions.

Measured calendar loading

ORM rows materialized

Isolated regression fixture: one relevant interval with unrelated calendars present. Loading fell from eight to three ORM rows and six to at most three SELECTs. Empty participant sets load no ORM records. This is a work-count measurement, not a latency or scalability benchmark.

Verification and Remaining Limits

Every implementation phase passed complete local C++ and Python gates before its focused commit. Database tests used real PostgreSQL in owned disposable schemas; operational data was not migrated or reset.

Accepted regression gates

Checkpoint	C++ passed / failed	Python passed / failed / skipped
Original baseline	185 / 0	481 / 0 / 1
Phase 2 - cleanup	185 / 0	483 / 0 / 1
Phase 3 - C++ reuse	187 / 0	483 / 0 / 1
Phase 4 - gateway	187 / 0	485 / 0 / 1
Phase 5 - database	187 / 0	488 / 0 / 1
Phase 6 - frontend	187 / 0	488 / 0 / 1
Phase 7 - test safety	187 / 0	490 / 0 / 1
Phase 8 - final	187 / 0	490 / 0 / 1

Final Python total: 491 executed; the one Windows skip covers POSIX executable permissions. Meaningful coverage was retained; nine Python cases and two C++ cases were added relative to baseline.

Integration evidence

Database: fresh Alembic head **0049b**, all **53 application tables**, no schema drift; real backup/restore, restart, rollback, identity, publication and concurrency checks passed. No schema migration was introduced.

Compatibility: all five allocation strategies, JSON/legacy CSV parity, all request types, appointments, room and equipment lifecycles, waiting lists, recurrence, feedback, calendars and notifications passed the complete suites. Standalone CSV CLI and optional import remain supported.

Frontend: 52 authenticated desktop/mobile page views across four roles; eight sample captures and computed layout metrics matched. Three additional real browser flows verified appointment, room and equipment submission, approval and cancellation, including private appointment denial and reservation-versus-handover distinction.

Static checks: 57 templates compiled, six JavaScript assets and 123 Python files passed syntax checks; schema diagrams, CI syntax and installed dependency consistency passed. Protected CSV hashes remained unchanged.

Reliability and deliberate limits

An experimental four-worker mixed-suite run previously timed out. Missing failure cleanup was corrected and guarded; four repeated recurring-workflow runs with two simultaneous module processes passed all 100 checks. The historical timeout cause remains unproven. Canonical serial test execution, which still tests real concurrent actors, remains supported.

Best-fit storage still grows with total candidate capacity. Large services and remaining legacy CSS are retained where a rewrite had no demonstrated benefit. No new framework, speculative schema, domain policy, deployment or allocation algorithm was introduced. Existing Strategy/Gateway/Adapter boundaries are reused, not multiplied.

Focused code commits

ea84cc8 cleanup; 483a183 best-fit reuse; 7b0ad3d gateway headers;
5a74ca7 scoped calendar reads; 24cb64f final selectors; 8e95aec failure-path tests.

Final Pre-Merge Safety Review

This page supplements the original phase gates on pages 1-2. The totals below are the latest post-correction verification; no feature, database schema, allocation objective or JSON version changed.

Investigated findings

FIXED - Best-fit arithmetic and resource safety. Positive individual capacities did not bound their sum or DP size. Checked addition now protects int and state-count arithmetic. Selection enforces a conservative 64 MiB allocation estimate and 50 million decision cells before DP allocation. Objective, candidate order and deterministic ties stay unchanged.

FIXED - Controlled failure at the execution boundary. Oversized work returns the existing JSON ENGINE_FAILURE / exit 3, never a false capacity rejection or a greedy fallback. CSV CLI catches allocation exceptions, exits 1 and does not replace exports. Real integration regressions verify both best-fit strategies and unchanged prior outputs.

VERIFIED SAFE - Worker cleanup and calendar semantics. Timeout and partial-start fault injection reaped all three real spawned workers, with no owned children left alive. Three focused real-PostgreSQL calendar tests passed. Actor scope, required membership, explicit academic periods, out-of-horizon bindings and unresolved dates remain intact; no calendar edit was needed.

DEFERRED - Historical experimental parallel timeout. The old four-worker mixed-suite stall remains unreproduced, with an unproven cause. Investigation is closed for this iteration as non-blocking. Supported serial execution and its actual concurrency tests pass; unrestricted full-suite parallel reliability is not claimed.

Bounded performance and memory sanity checks

Candidates x capacity	Selected rooms	Elapsed (ms)	Peak resident (MiB)
10 x 100	5	0.04	4.2
100 x 100	50	1.46	4.5
250 x 200	125	15.16	6.4
5 x 800000	3	33.09	38.3

Optimized helper-only Windows probes with concurrent verification load; not a production benchmark or SLA. The unsafe pre-fix probe denied an 80,000,008-byte allocation attempt before memory was consumed; the corrected guard refuses before DP allocation.

Accepted full verification and remaining limits

C++: 192 passed / 0 failed, plus both isolated CLI smokes. **Python:** 493 executed, 492 passed / 0 failed / 1 Windows POSIX-permission skip. All five strategy parity cases, real PostgreSQL, migrations, rollback, concurrency and recovery are included. Head remains **0049b**; diagrams match all **53 tables**. Targeted gateway: 24 passed. Correction commit: **168c55f**.

DP cost still scales with capacity. Mathematically valid inputs above the technical budget fail explicitly. The estimate is not a whole-process RSS limit and cannot prevent external system-wide memory exhaustion. No operational database was migrated/reset; no policy, permission or calendar behavior changed. The branch remains for review, not merged or deployed.